

We Need More Engineers with PhDs! And...

Opportunities for Engineering Research in Ocean Science

Matthew J. Rau

Nov. 14th, 2025

matthew.rau@gwu.edu



 **Why you should put grad school on your radar**

 **How I got here**

 **Why you should care about organic ocean particles**

 **Things my lab is working on to understand these better**

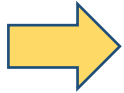
 **Bio-inspired efficient particle filtration mechanisms**



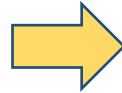
First, a challenge: What do you want to be doing 5 years AFTER you graduate? 10 years? 20 years?

Education was explained to me this way...

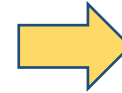
School



University

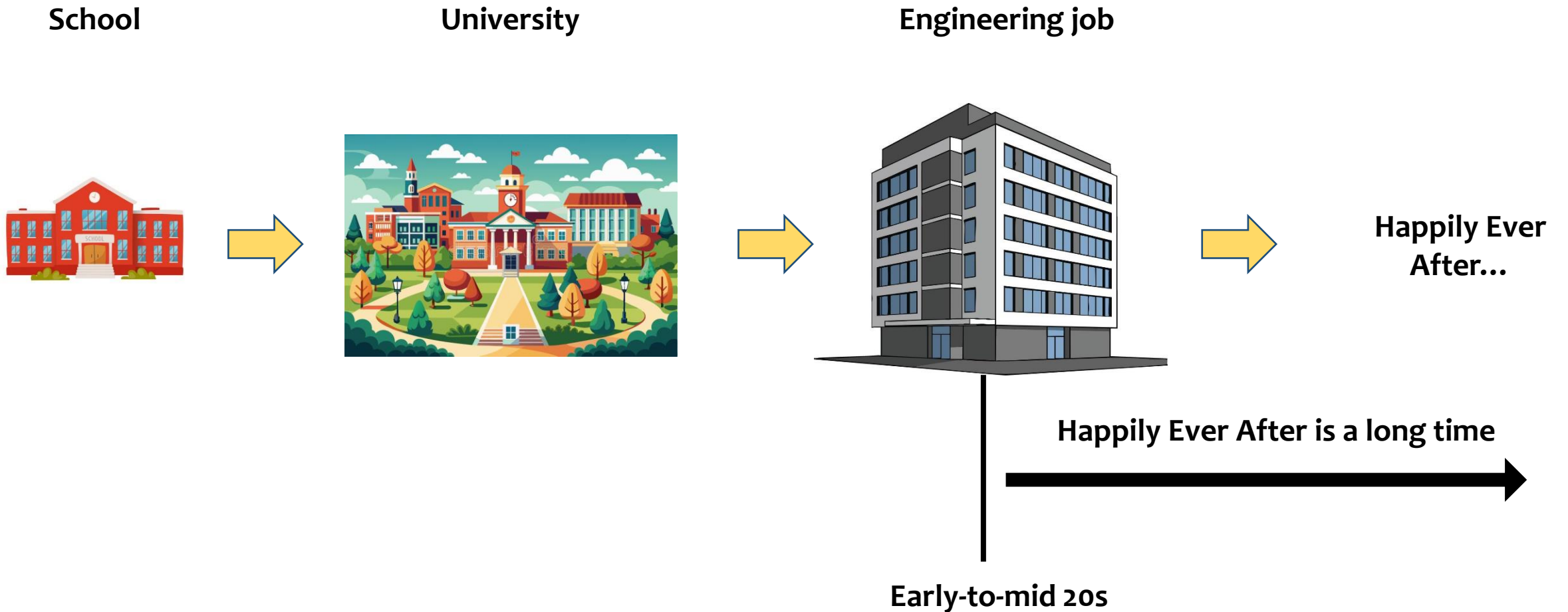


Engineering job

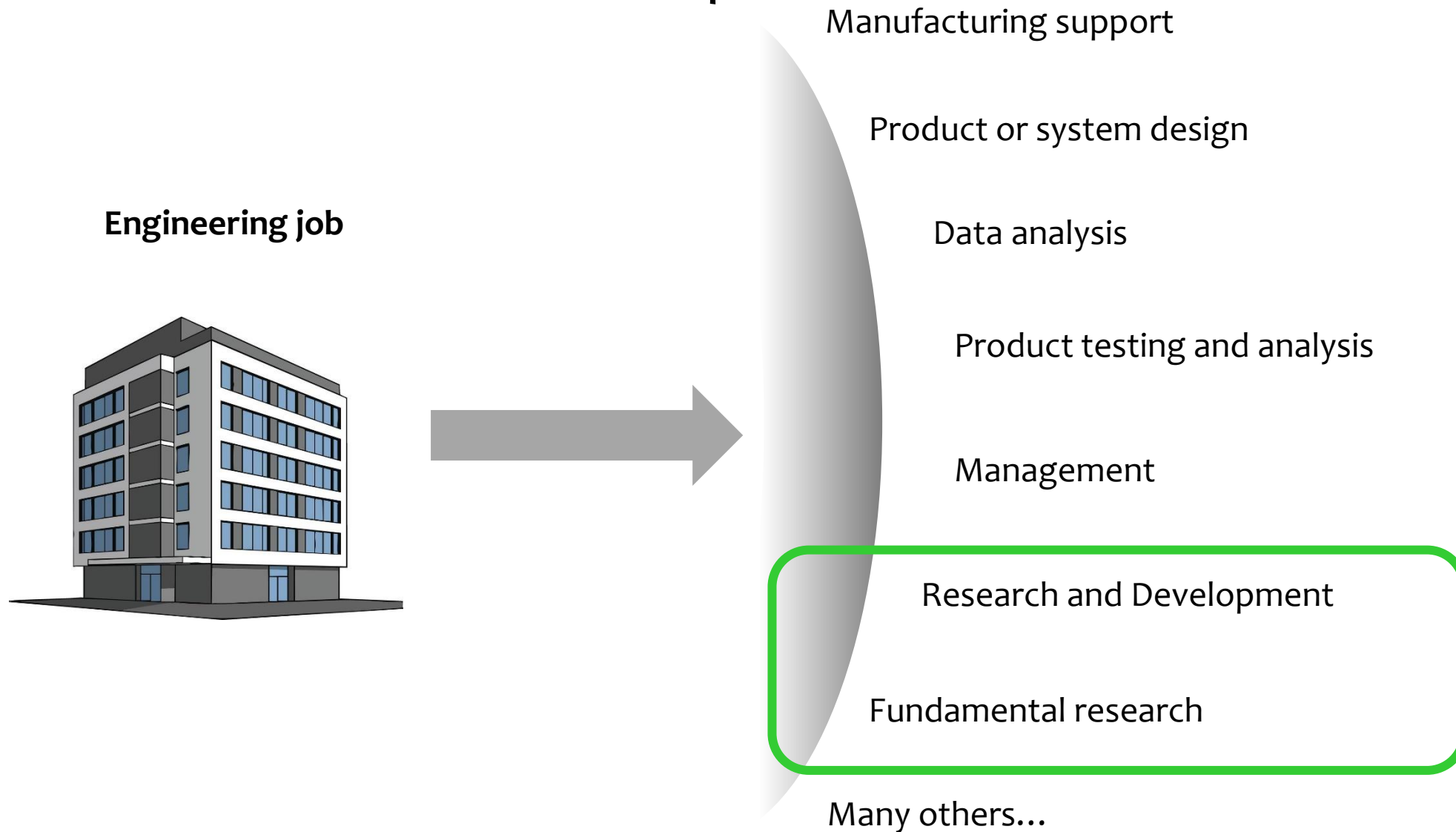


**Happily Ever
After...**

Education was explained to me this way...

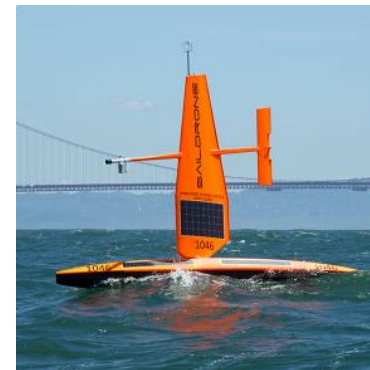
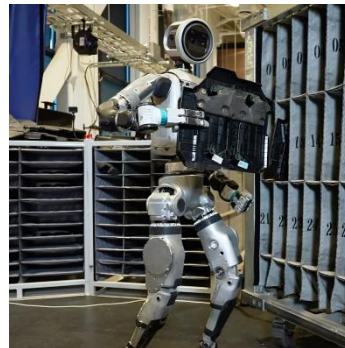
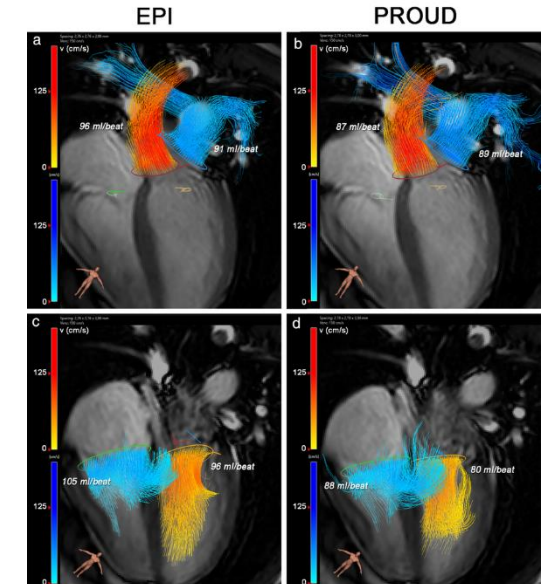


You can take many paths as an ME and even change directions mid-career – MEs have options!

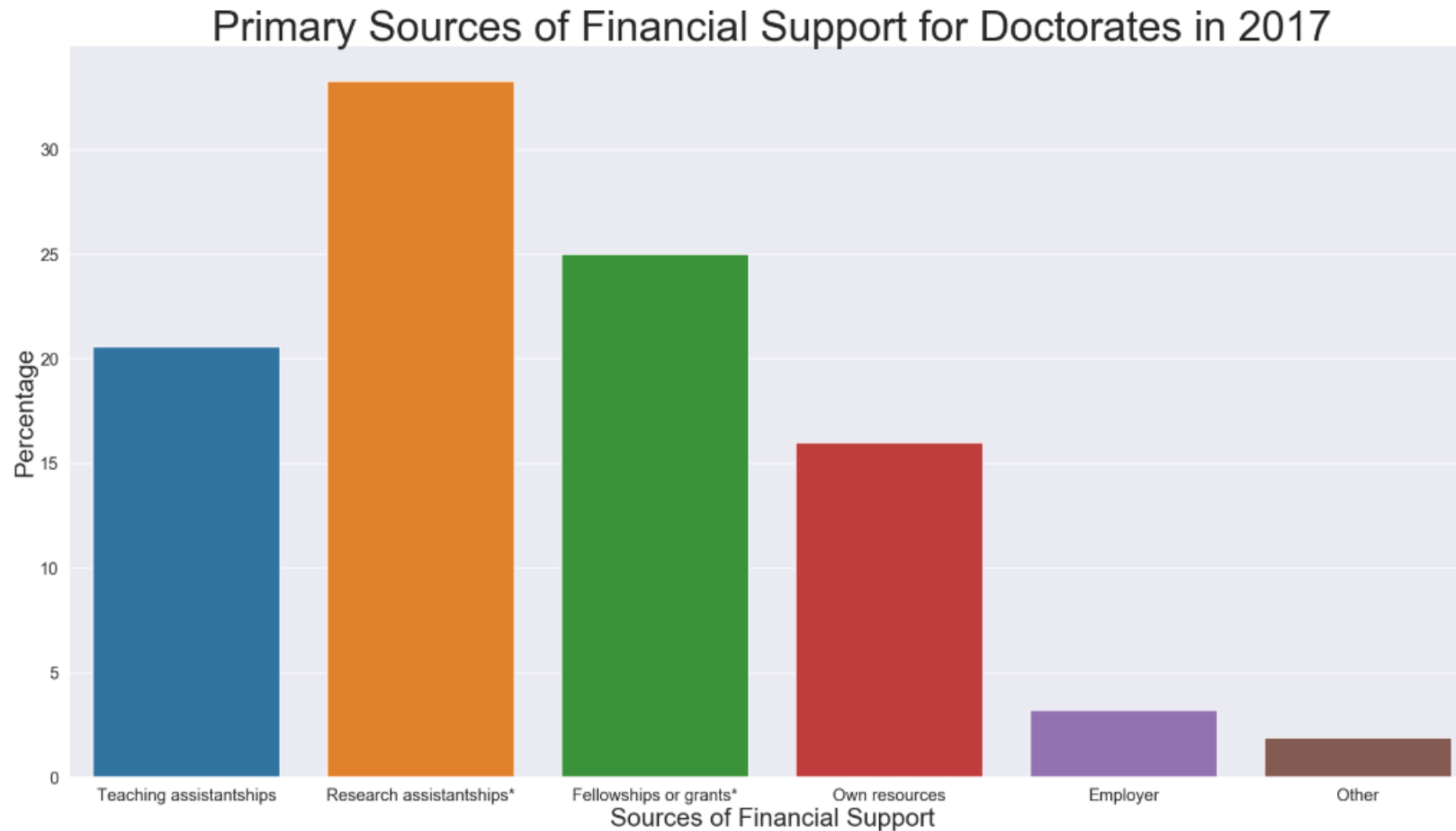


A case for thinking about graduate school, especially a PhD

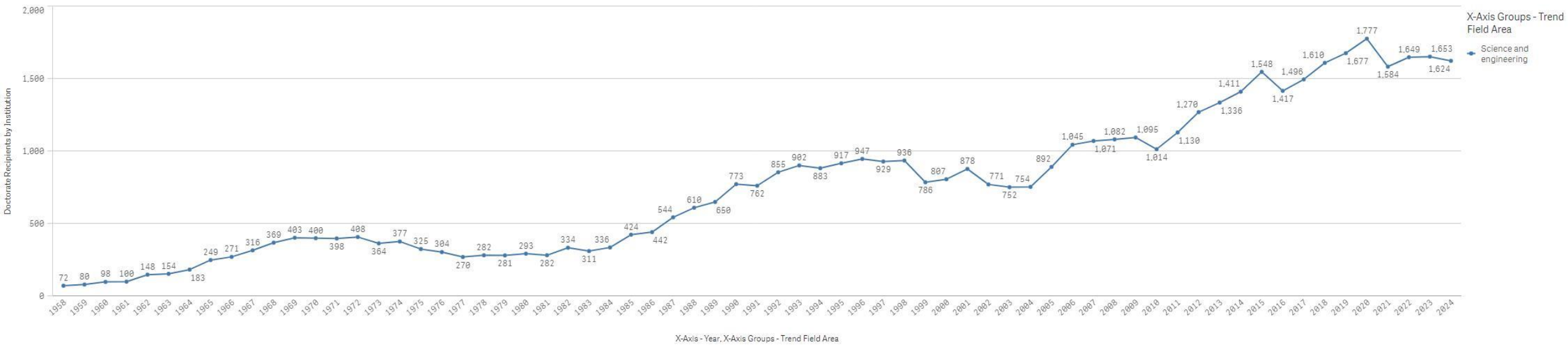
- Someone will very likely PAY YOU to continue your education
- You learn more about the subjects you like the most
- You'll be well compensated (after grad school)
- You will get to then build a career in Research and Development
 - Every day is different!



Often, you are paid to do research with a faculty member as part of your education/training

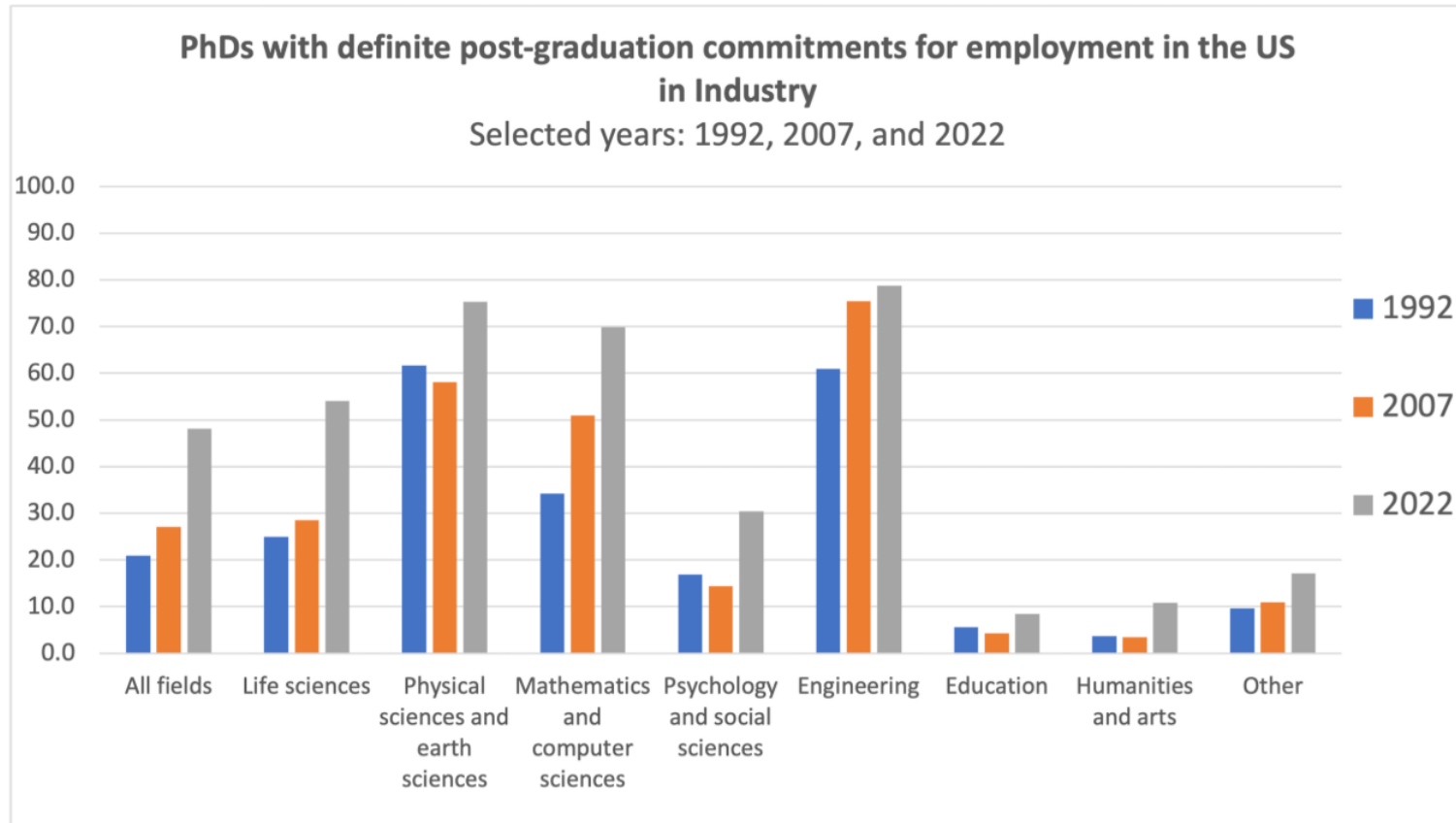


There aren't that many ME PhDs... We need more!



SOURCE: National Center for Science and Engineering Statistics, Survey of Earned Doctorates.

Don't use your Profs as examples – most PhDs go to work in industry and do exciting R&D



Qualcomm



SPACEX



TOYOTA



Pratt & Whitney
A United Technologies Company

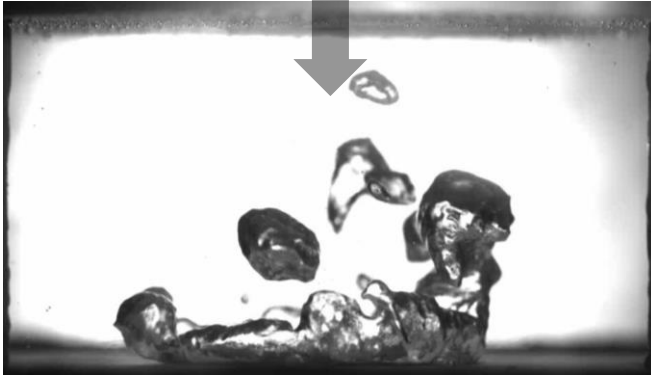


GE Aviation

Basically every company that does R&D wants to hire engineers with PhDs.

My engineering path was not obvious

$Re = 5,000$ Jet $T_{jet} = 90\text{ }^{\circ}\text{C}$
 $T_{sat} = 100\text{ }^{\circ}\text{C}$



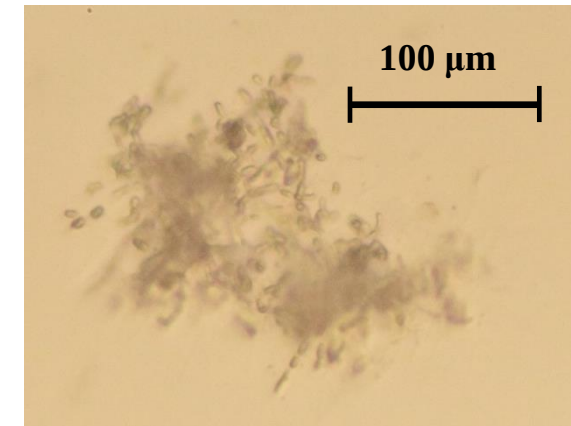
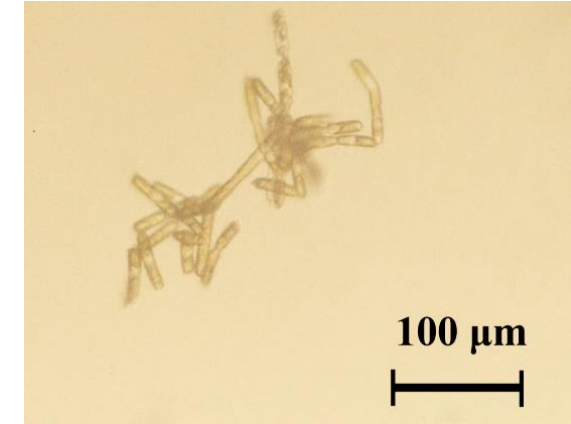
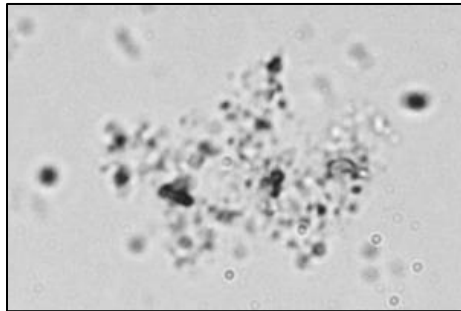
U.S. NAVAL
RESEARCH
LABORATORY



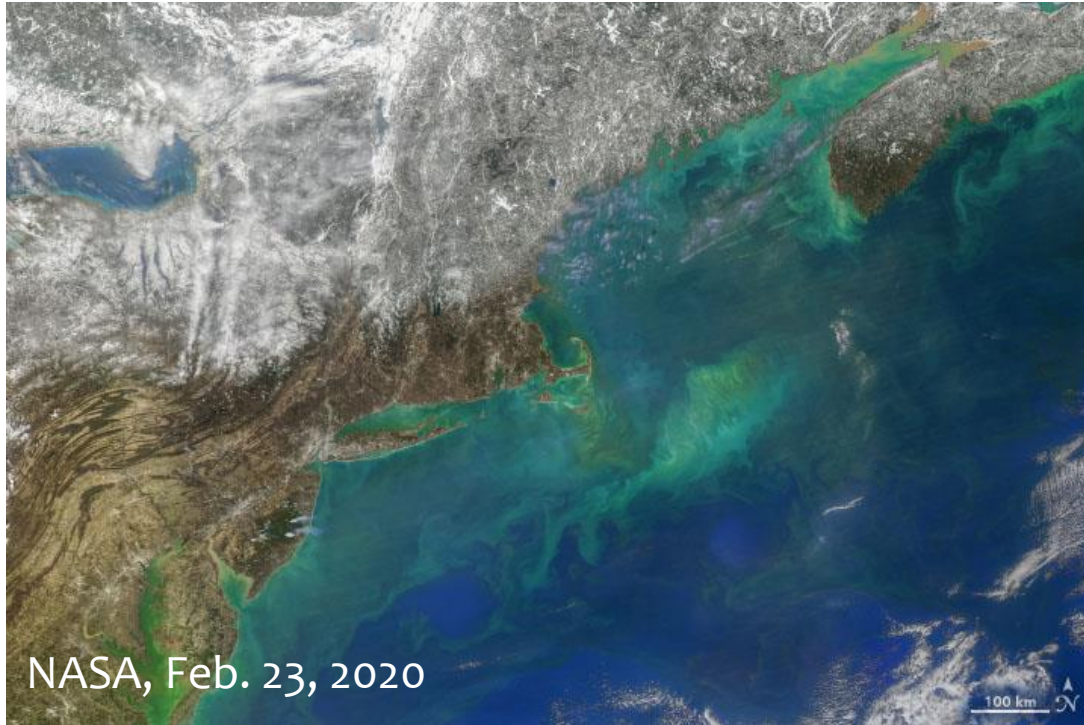
TOYOTA

↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑
Heated area

example clay floc



Phytoplankton forms much of the particulate matter in the ocean



The New York Times

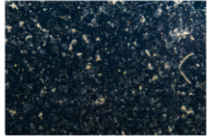
SCIENCE

In the Ocean, It's Snowing Microplastics

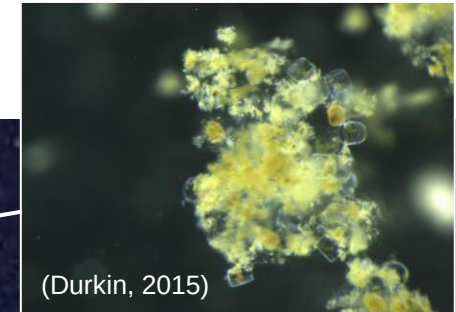
Tiny bits of plastic have infiltrated the deep sea's main food source and could alter the ocean's role in one of Earth's ancient cooling processes, scientists say.

By Sabrina Imbler

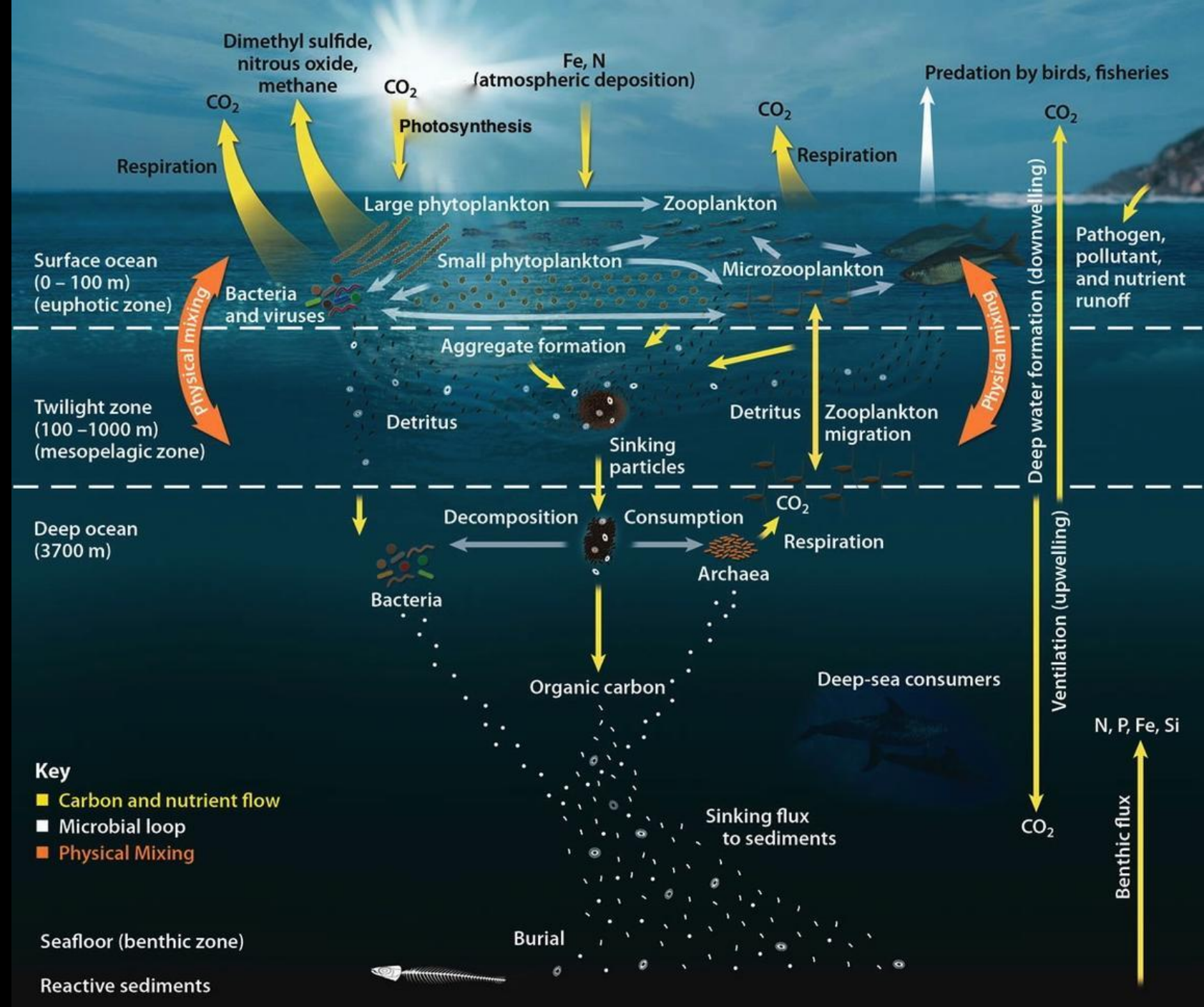
PRINT EDITION Deep-Sea Disruption | April 5, 2022, Page D1



Marine snow

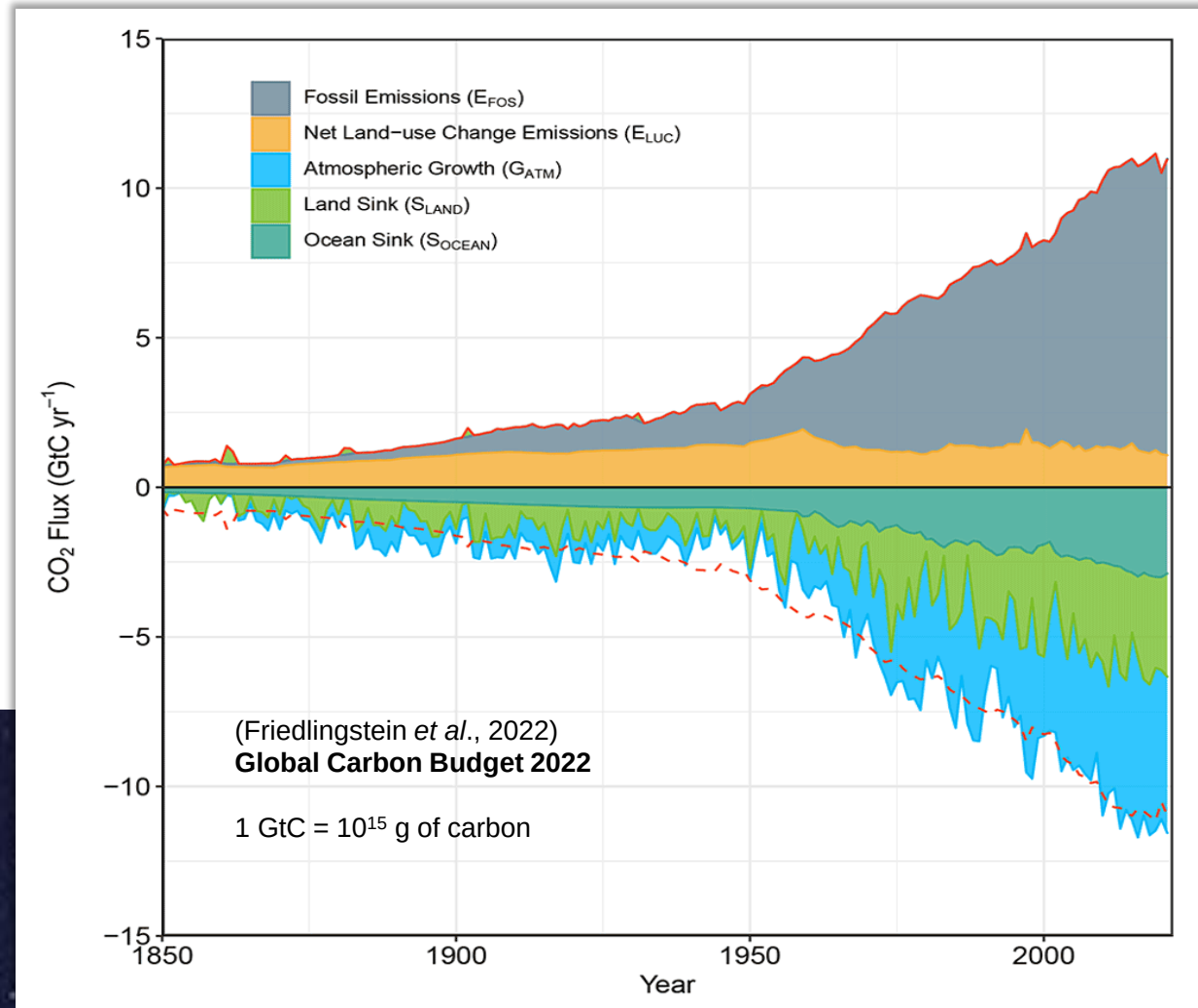
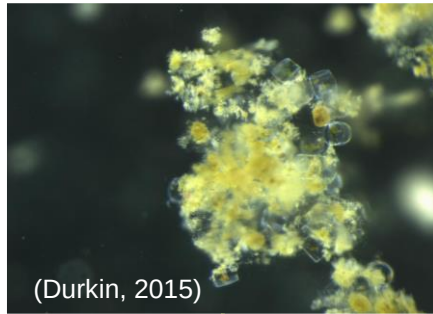


My lab is focusing on how particles move and change due to the dynamics of the turbulent ocean



The settling of marine snow in the ocean is a key aspect of the marine carbon cycle

Marine snow



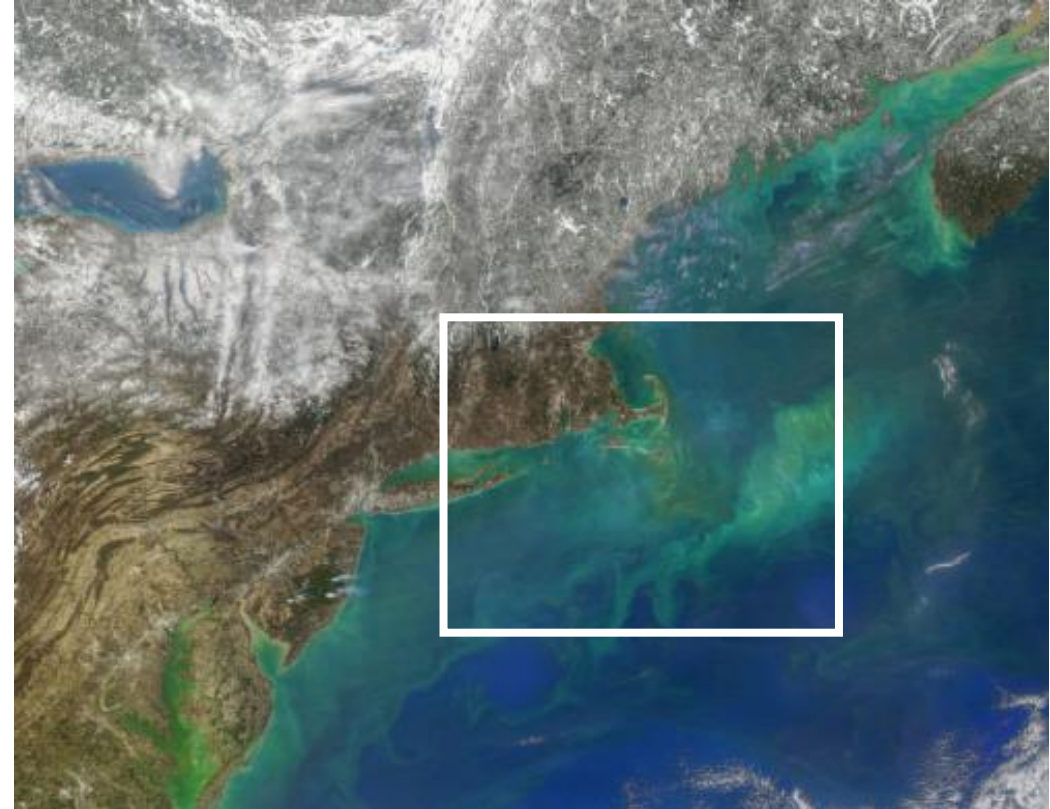
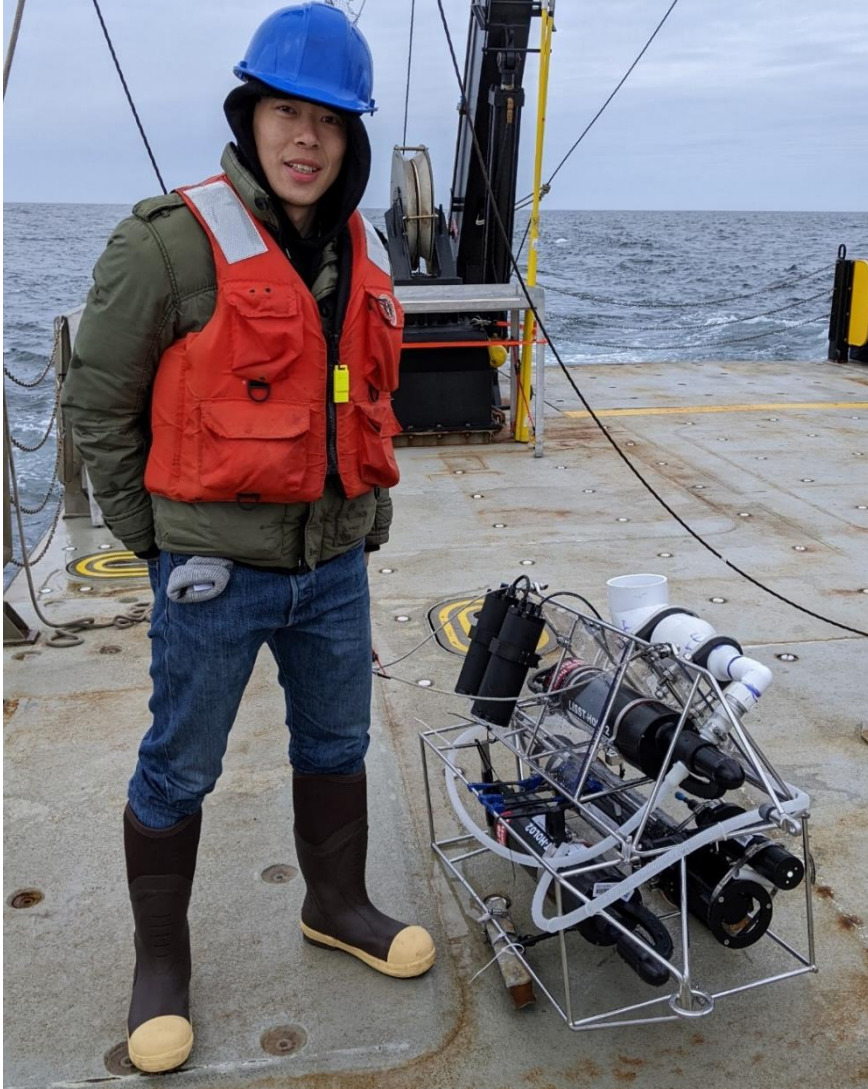
My lab does a variety of experimental and field measurements to understand the dynamics of marine phytoplankton aggregates



0.625 g/L XG



We also take field measurements and build custom equipment to make unique particle characterizations



We measured turbulence as part of our campaign to link fluid flow to particle properties

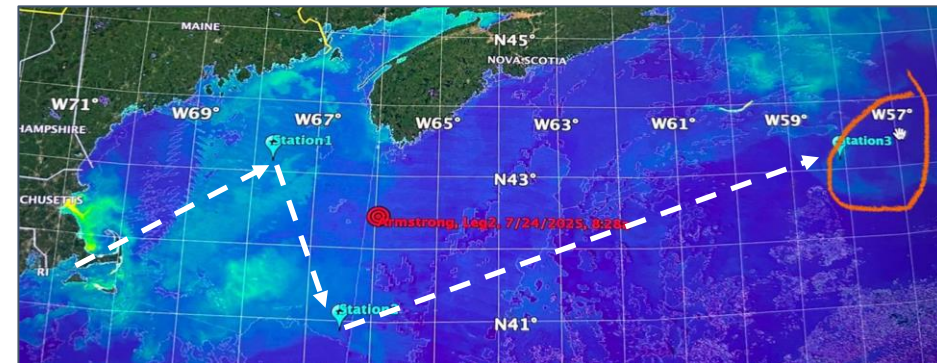
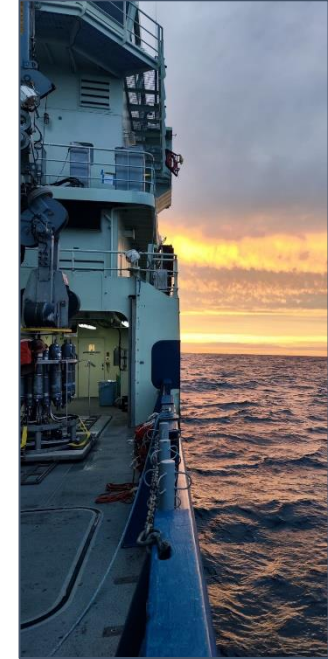


We are currently investigating the role of biological responses to the environment (including turbulence) in the formation of aggregates

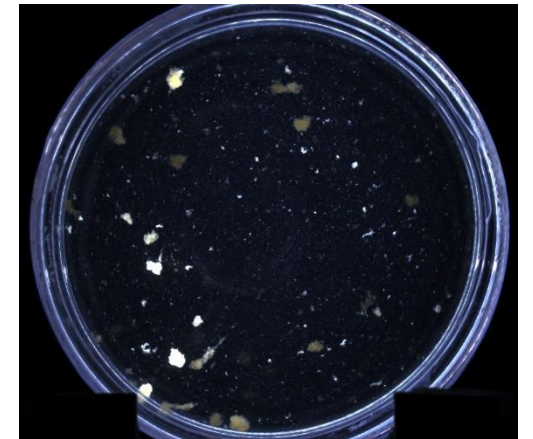
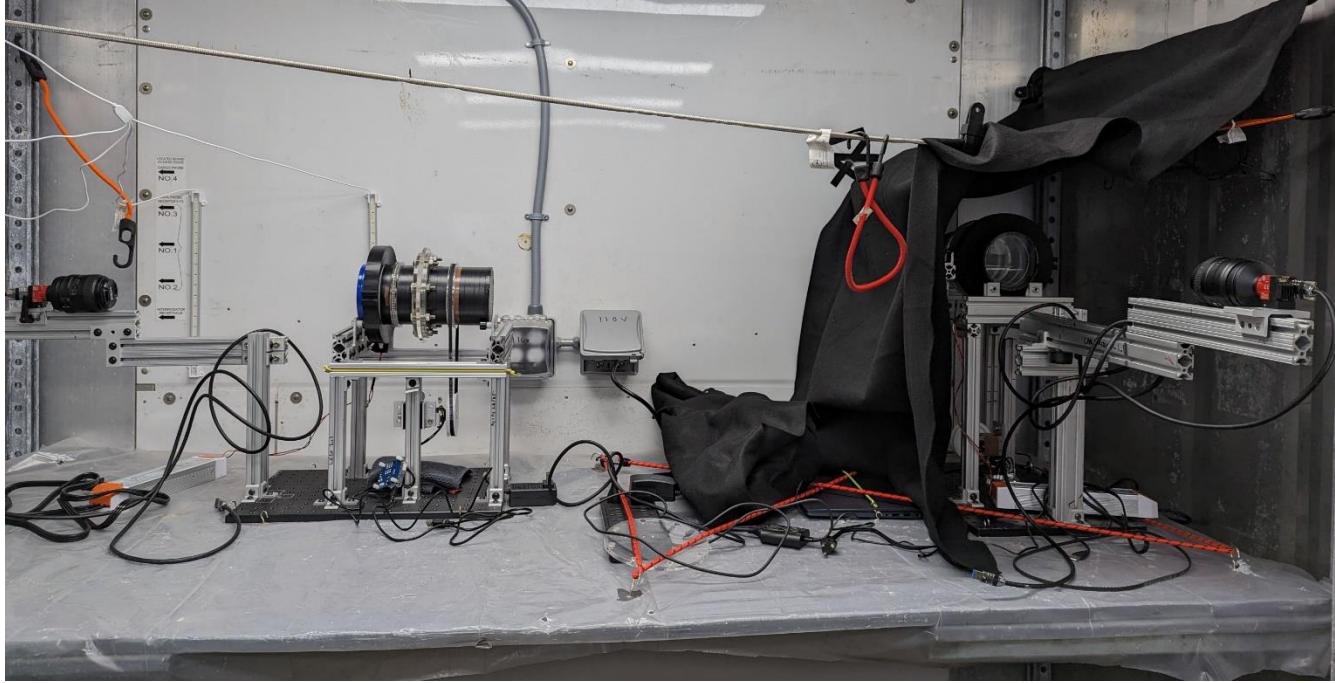
VIISTA 1, June 2024



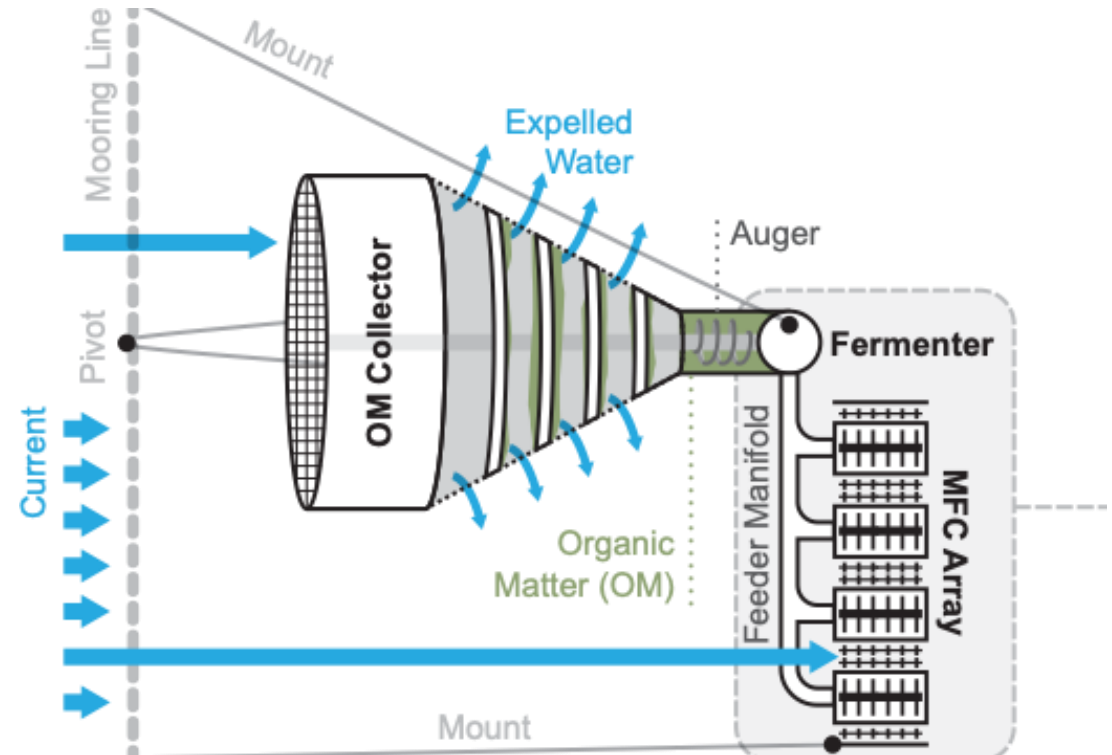
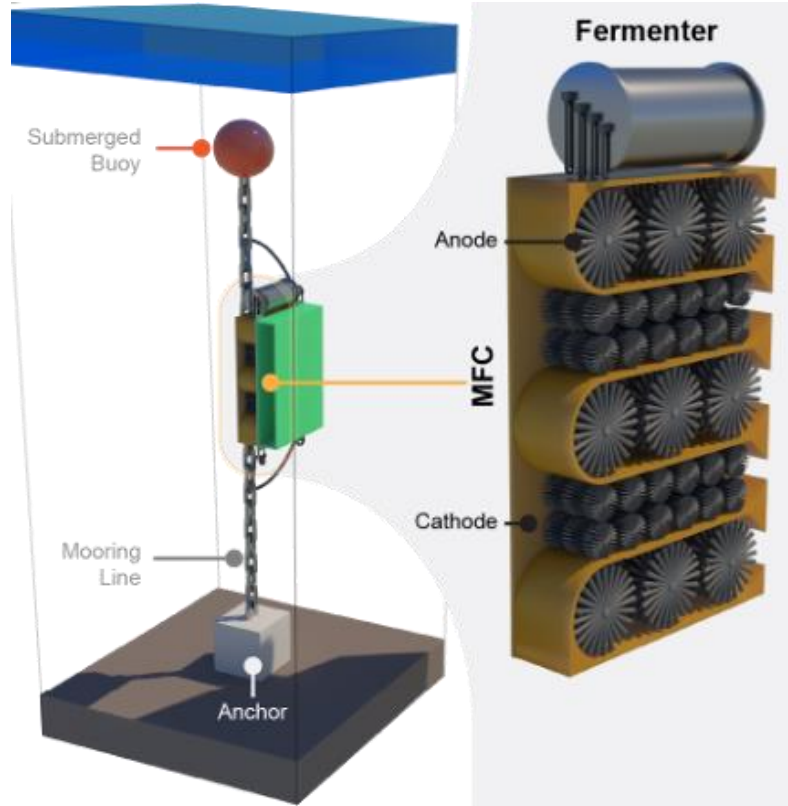
VIISTA 2, July 2025



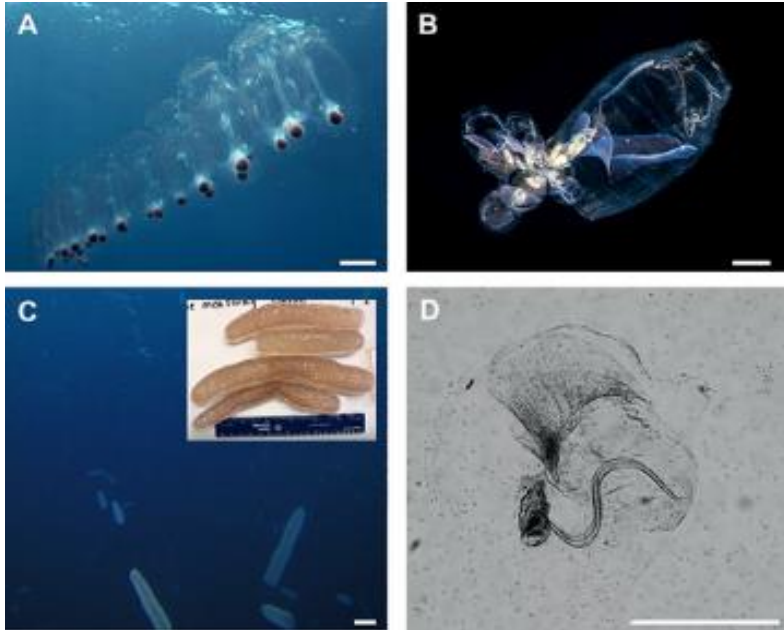
This has included additional aggregation experiments on ships...



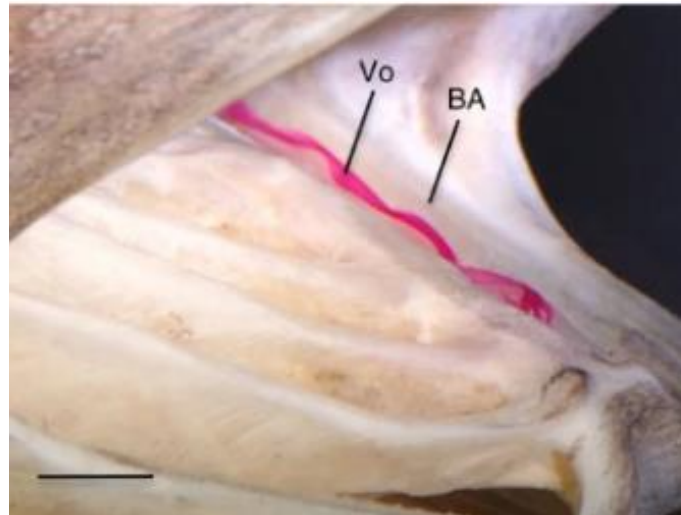
We are part of a team to develop a long-term oceanic power source submerged in the upper water column fueled by organic material



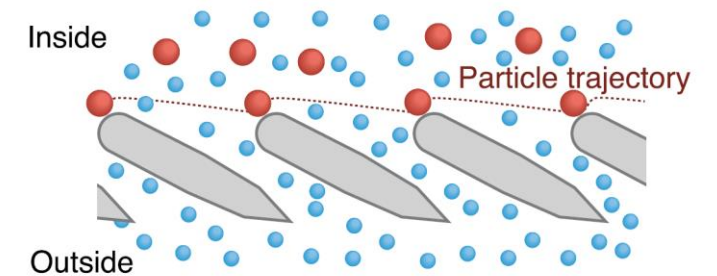
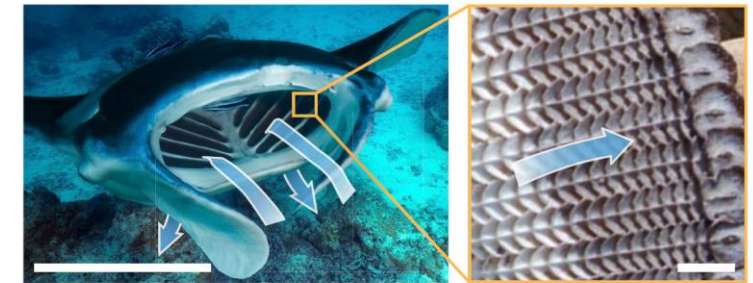
Nature has already developed efficient ocean particle filtration mechanisms



Sutherland and Thompson,
Limnol. Oceanogr, 2022.



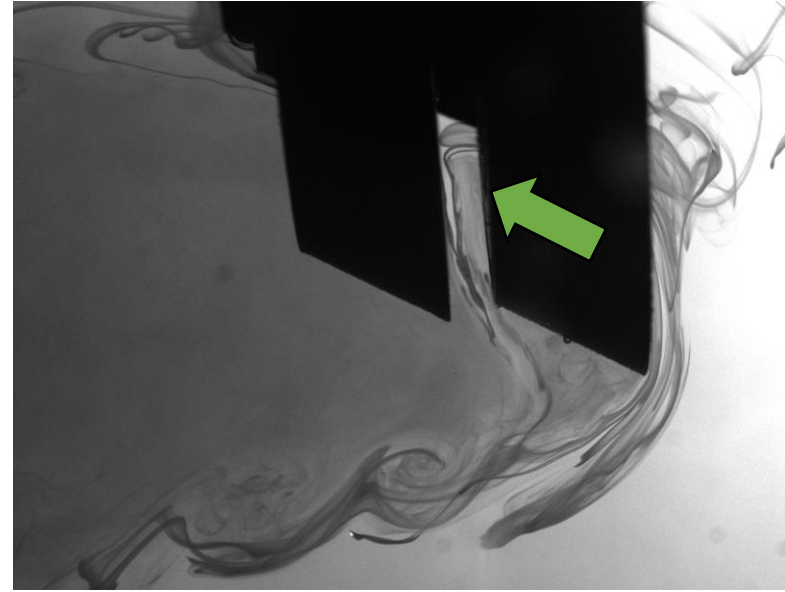
Sanderson et al., *Nature Communications*, 2018.



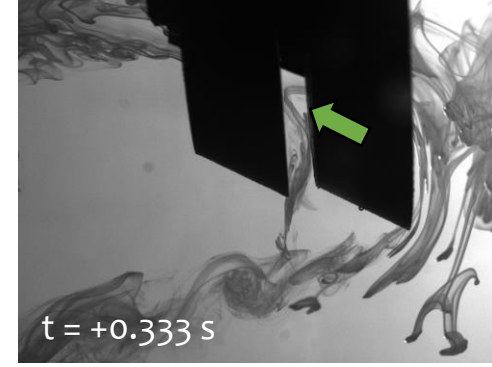
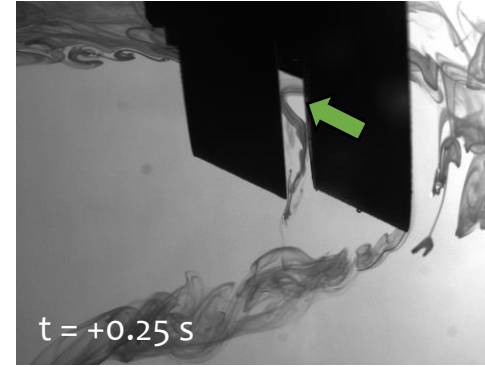
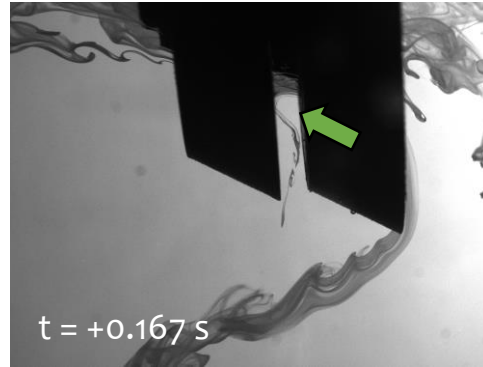
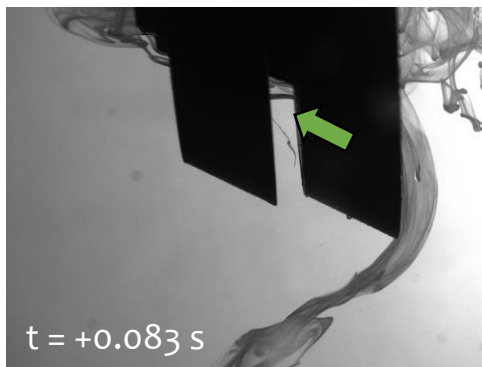
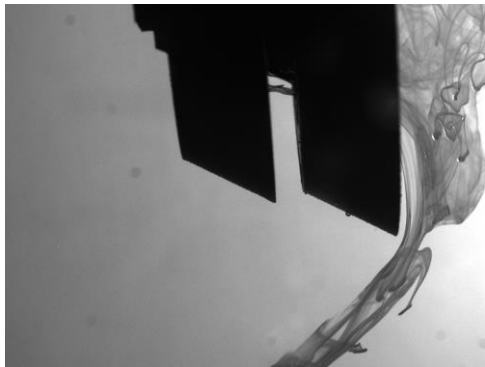
Mobula filtration

Mao, et. al (2024), *PNAS*, 121(50),
e2410018121.

Inspired by this, we are designing 'structured' filter lobes that leverage vortex formation to promote particle capture



Freestream
velocity of
 ~ 0.1 m/s



We track particles in flume experiments and perform CFD analysis to quantify design performance

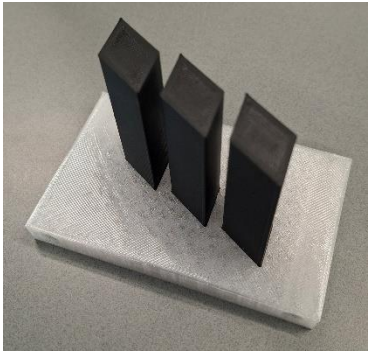
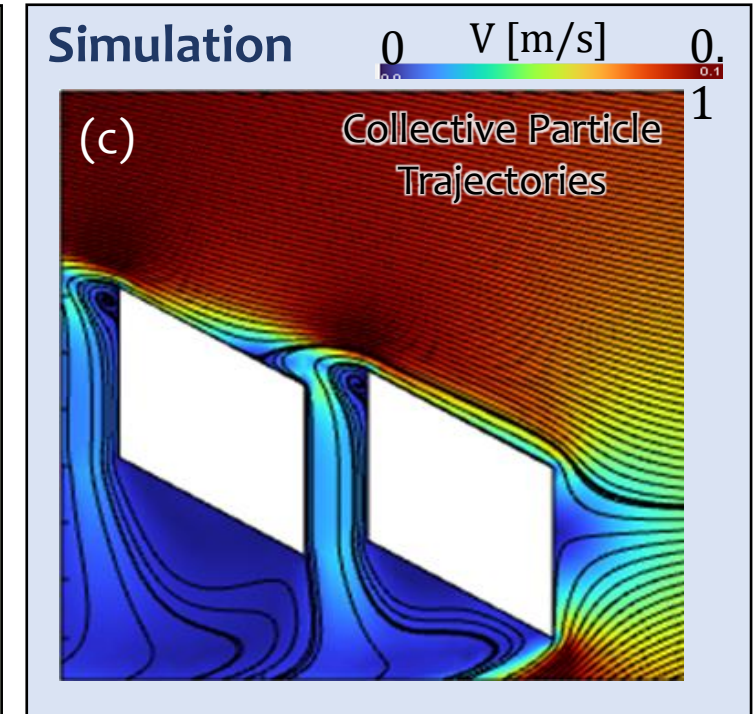
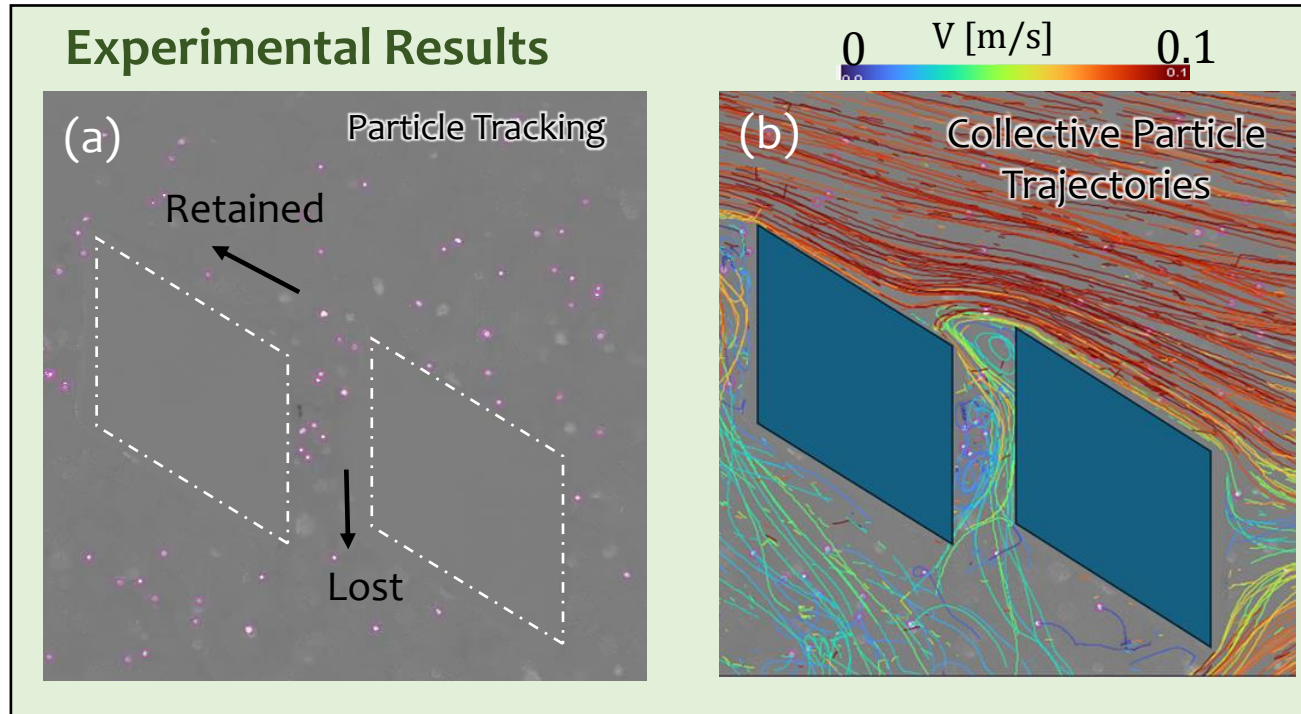
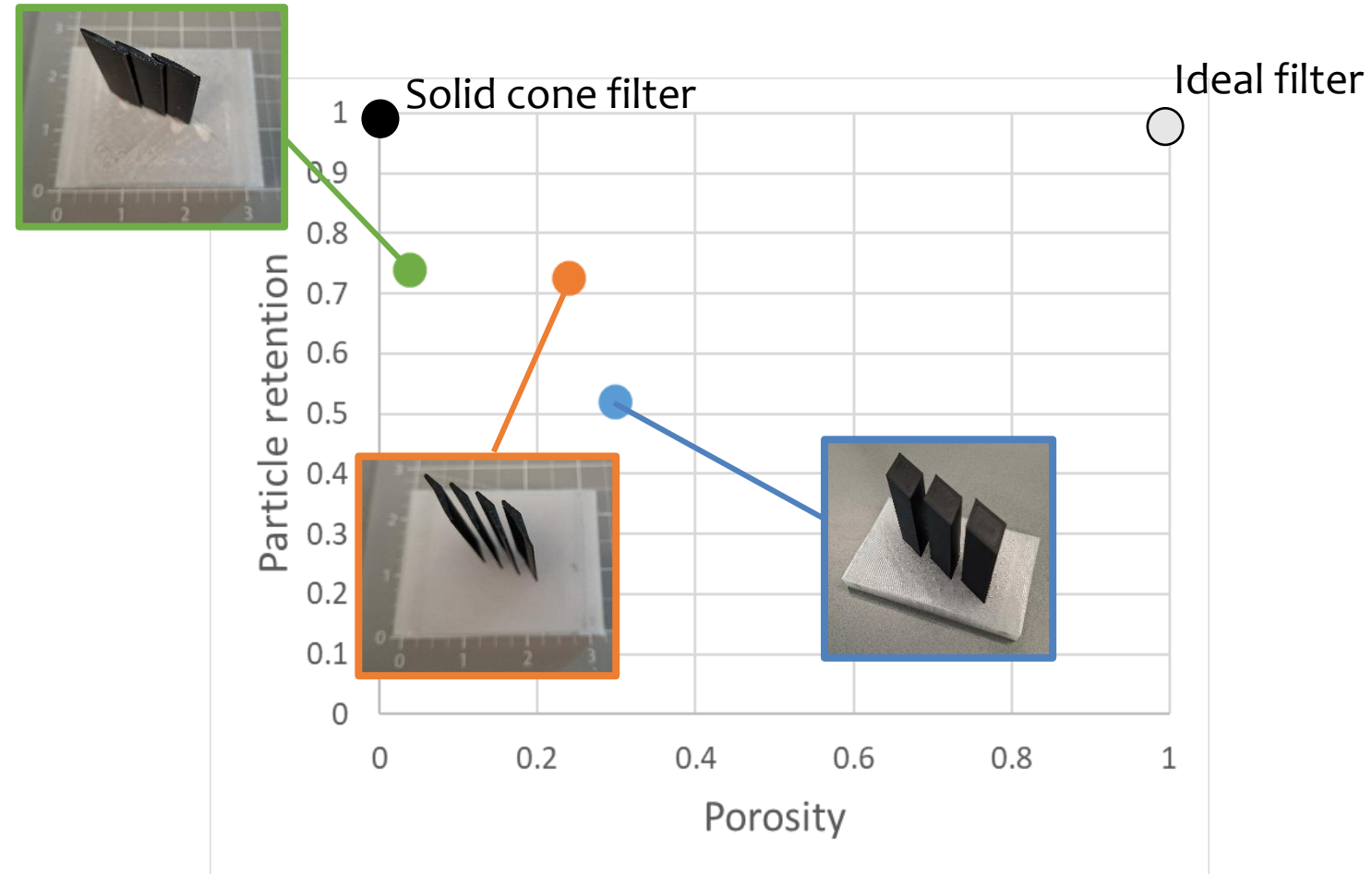


Figure: 3D printed rib test coupon



Rib shape is a critical design parameter for achieving a tradeoff between particle capture and filter permeability

	SD002	Airfoil	Mobula Ray
Particle Retention ϵ [%]	52	74	72
Porosity [-]	0.3	0.04	0.24



Field testing confirms the ability of these designs to capture real oceanic organic particles



ADCP deployed up river of dock
Average current speed ~ 0.24 m/s

In Conclusion

Put graduate school
on your radar!



matthew.rau@gwu.edu

Ask faculty if you can work
in their research labs! This
is great experience.



Keep an open mind. You never
know what research topics are
important problems

